SCOOT · Level 1 · Foundation

The traffic light that sees the queue before it forms

How SCOOT counts cars far up the road, predicts the queue at the lights, and nudges the timings a few seconds at a time.

6 ideas, each building on the last

1 What SCOOT is: the elastic plan

~192 small timing decisions per hour at one junction

2 Detectors and link profile units

1.96 vs 3.07 LPUs for a car and a bus at 13.5 m/s

3 Profiles and platoon dispersion

F = 0.263 smoothing factor on a 10-second link

4 The queue model and degree of saturation

90% SCOOT's usual degree-of-saturation target

5 Validation: making the model match the street

4 s clear-time error from a 12% SATO error

6 SCOOT against fixed time and actuation

12.8 s SCOOT mean delay in Lab F3 (fixed 19.0 s)

Three surprises to start

The detector is in the 'wrong' place on purpose.

Most actuated signals put their loops at the stop line. SCOOT puts them near the far end of the road, just after the previous junction. That gives it 10 to 20 seconds of warning: it knows a platoon is coming before any driver reaches the lights.

SCOOT never sees the queue.

There is no camera watching the queue. Every queue SCOOT uses is a prediction from its model. In our SUMO lab the model's queue still tracked the real queue closely, second by second.

It has been running since about 1980.

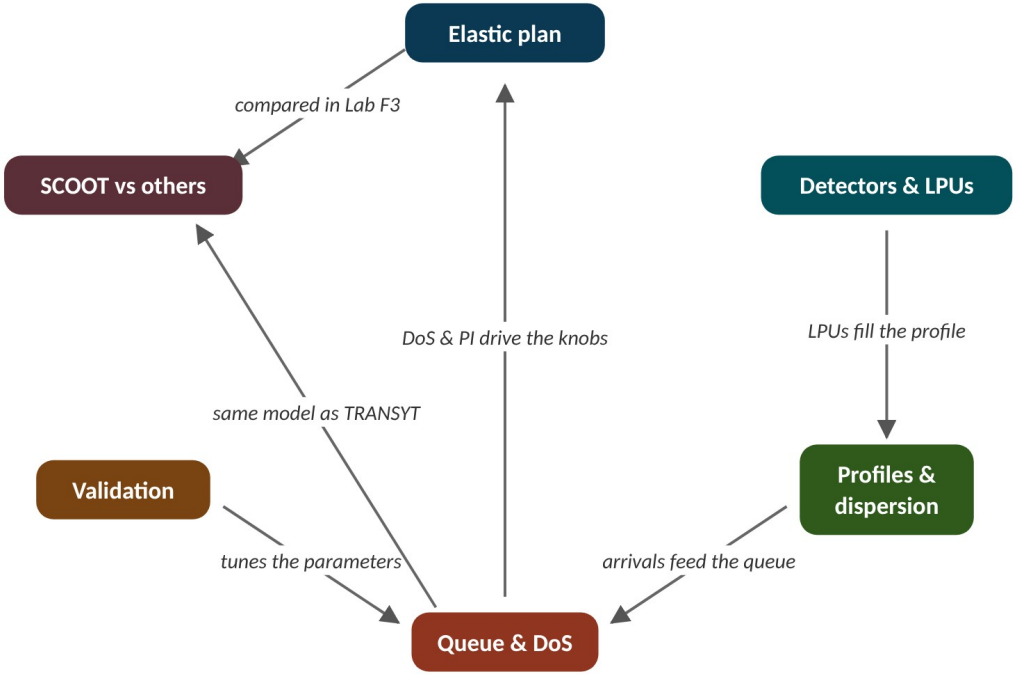
SCOOT was developed in the UK by the Transport and Road Research Laboratory (now TRL) and is used in hundreds of towns and cities. SCOOT 7 (2020) added connected-vehicle data and pedestrian features.

Words you will meet

Junction / node	A place where roads meet and traffic lights take turns. SCOOT calls each one a **node**.
Link	One approach road into a junction, from the detector to the stop line.
Stage	A set of movements that get green at the same time, for example 'main road both ways'.
Cycle	One full turn through all stages, for example 60 seconds.
Split	How the cycle is shared between stages: 'main road 35 s, side road 20 s'.

Offset	When one junction's cycle starts compared with its neighbour, so a green wave can form.
Platoon	A bunch of cars travelling together, usually released by the previous green.
Detector (loop)	A wire loop in the road that knows when a vehicle is over it.
Saturation flow	The fastest a queue can leave on green, roughly one car every two seconds per lane.

The map of today's ideas



Your mission: investigate in groups

1 Get your question

Six groups, one question each. Read it together and watch its animation twice.

2 Investigate (15 min)

Start with the fundamentals marked for your group (F1–F9), then your section's slides, the study guide and the website. Do the small calculation by hand.

3 Report back (3 min)

One sentence answer, one key number, and explain the animation to the class in your own words.

4 Discuss

The class asks one question to each group. Then we check against the answer slide.

THE SIX QUESTIONS

- **Group 1** Small nudges
- **Group 2** Detectors & LPUs
- **Group 3** Platoons spread
- **Group 4** Queue & DoS
- **Group 5** Validation
- **Group 6** SCOOT vs others

Fundamentals for every group

Nine short building blocks before you start. Each group needs only some of them: the chips on every slide tell you which.

F1 What a traffic signal controls

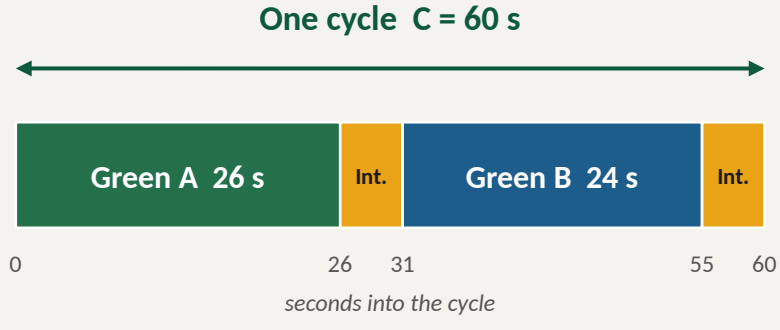
A signal shares one road space between movements that would collide. It gives each group of movements a turn, over and over, like a teacher letting one row of students out at a time.

› **Stage**: movements that get green together (for example the main road, then the side road).

› **Cycle C**: the time to run every stage once, for example 60 s.

› **Intergreen**: amber + all-red between stages; partly lost time, when nobody moves well.

› **Split**: how the green is shared, for example 26 s for the main road and 24 s for the side road.



Split A 26 s : B 24 s

Intergreen 2 × 5 s per cycle

TRY THE NUMBERS 60 s cycle = 26 s green A + 5 s intergreen + 24 s green B + 5 s intergreen.

FOR GROUPS 1 4 6

F2 A road network in SCOOT words

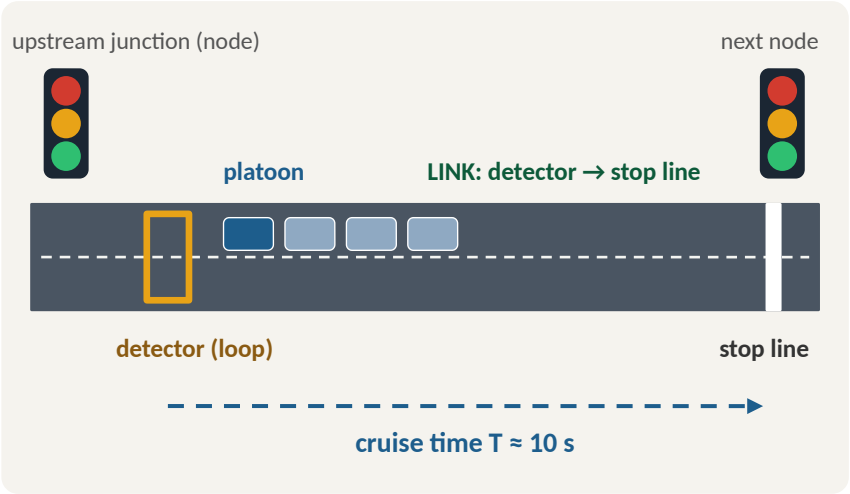
SCOOT sees a town as junctions joined by one-way approach roads. Each approach has its own detector and its own little model.

› **Node**: a signalled junction.

› **Link**: one approach road, from its detector to its stop line.

› **Platoon**: a bunch of cars released together by the upstream green.

› **Cruise time T**: how long a car takes from the detector to the stop line, for example 10 s.



TRY THE NUMBERS A 140 m link driven at 14 m/s (50 km/h) has a cruise time of 140 ÷ 14 = 10 s.

FOR GROUPS 2 3 5

F3 Measuring traffic: flow, headway, occupancy

Three ways to describe the same stream of cars: how many pass, how far apart they are in time, and how much of the time the detector is covered.

› **Flow q**: vehicles per hour (veh/h) or per second.

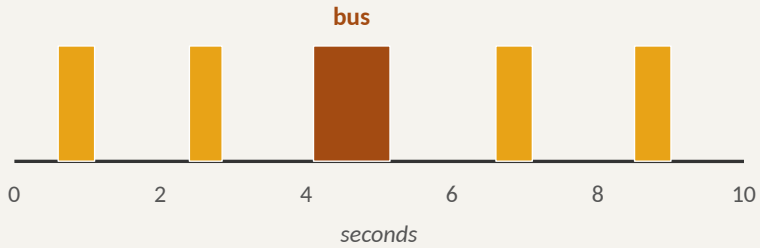
› **Headway**: seconds between one car and the next; headway = 1 ÷ flow.

› **Occupancy**: share of time the detector is covered. Long or slow vehicles cover it longer.

› SCOOT's **LPU** mixes the count with the covered time (group 2).

TRY THE NUMBERS 540 veh/h = 540 ÷ 3600 = 0.15 veh/s, so one car every 1 ÷ 0.15 ≈ 6.7 s.

Is the detector covered? (10 seconds)



5 vehicles in 10 s → flow 0.5 veh/s, average headway 2 s
covered 3.0 s of 10 s → occupancy 30%

FOR GROUPS 2 3 4

F4 Saturation flow and capacity

When the light turns green, a standing queue leaves at a steady top speed, one car about every 2 s per lane. That top rate is the saturation flow. SCOOT measures the same idea in LPU per second and calls it SATO.

› **Saturation flow s**: about 1800 veh/h per lane (1 car every 2 s) on a good lane.

› **Start lag**: the first cars take a few seconds to get going.

› **Capacity** = s × g ÷ C: what the approach can carry per hour.

› Narrow lanes, hills, parking and turning traffic lower s.



A queue drains at the saturation rate (shown in LPU/s)

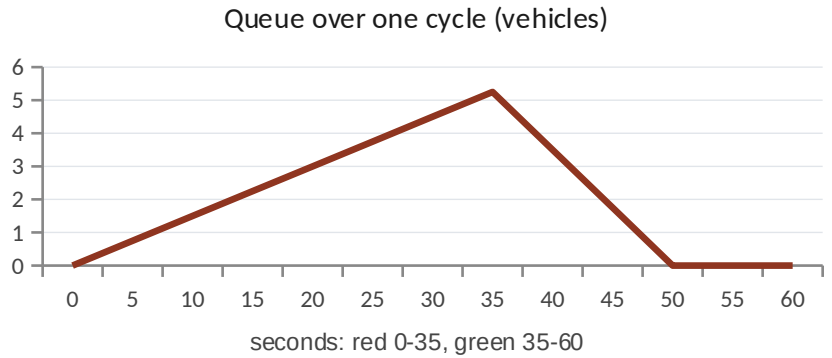
TRY THE NUMBERS s = 1800 veh/h, green g = 25 s, cycle C = 60 s: capacity = 1800 × 25 ÷ 60 = 750 veh/h.

FOR GROUPS 4 5

F5 Queues, delay and stops

During red, arriving cars pile up. During green the queue drains faster than cars arrive, so it shrinks. Waiting time is delay; every car that had to brake to a halt is a stop.

- Queue grows in red at the **arrival rate**.
- In green it shrinks at **saturation flow − arrival rate**.
- **Delay** = area under the queue line (vehicle-seconds).
- **Stops** = cars arriving before the queue clears.



area under the line = delay ½ × 50 × 5.25 ≈ 131 veh·s

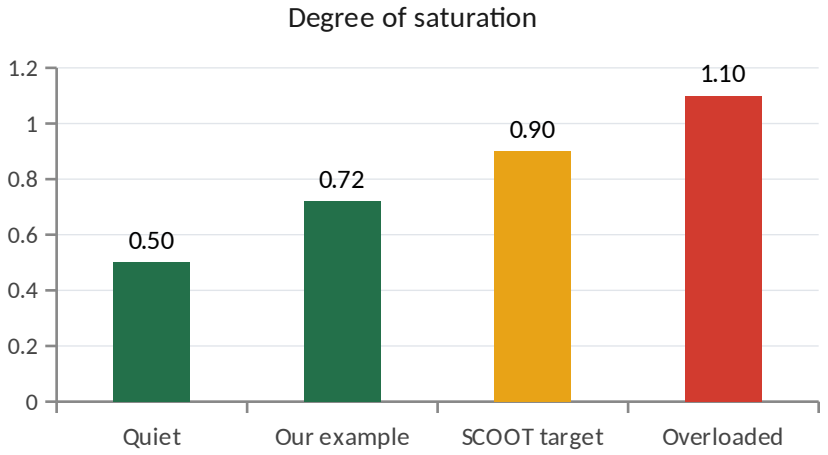
TRY THE NUMBERS Arrivals 0.15 veh/s, red 35 s, s = 0.5 veh/s: queue 0.15 × 35 = 5.25 veh, clears after 5.25 ÷ 0.35 = 15 s of green.

FOR GROUPS 4 6

F6 Degree of saturation: is the green big enough?

Compare what arrives with what the green can clear. A number below 1 means spare green; above 1 means the queue grows every cycle.

- **DoS** = demand ÷ capacity.
- Below about **0.9**: comfortable, room for random surges.
- About **1.0**: the green is just used up.
- Above **1.0**: queues grow every cycle, whatever the timing.



TRY THE NUMBERS Demand 540 veh/h, capacity 750 veh/h: DoS = 540 ÷ 750 = 0.72 (72%).

FOR GROUPS 1 4 5

F7 Coordinating junctions: offsets and green waves

Neighbouring signals can be timed so a platoon released by one arrives at the next just as it turns green. The time between their greens is the offset.

› **Offset**: time from green at one junction to green at the next.

› Good offset ≈ travel time between them, so the platoon meets a green.

› Neighbours must share a cycle (a **region**) to keep the wave.

› The wave works best when platoons stay tight (group 3).



Trying offsets to catch the platoon on green

TRY THE NUMBERS Junctions 300 m apart at 15 m/s: travel time 20 s, so set the second green about 20 s after the first.

FOR GROUPS 1 3

F8 Three ways to run a signal

All signals share the road; they differ in how they decide the green.

Fixed time

Runs a stored plan, often made off-line with Webster or TRANSYT. Cheap and predictable, but cannot react.

Actuated

Loops at the stop line extend the green while cars keep coming, and end it on a gap. Reacts, but only locally.

Adaptive (SCOOT)

Loops upstream feed a model; small, frequent changes to split, cycle and offset across a region.

TRY THE NUMBERS Rule of thumb: the more traffic changes and the closer the junctions, the more an adaptive system can help.

FOR GROUPS 1 6

F9 Your investigation toolkit

Every group follows the same four moves. The table shows which building blocks to revise first.

- **Watch** your animation twice: what changes, and why?
- **Explain** it in everyday words, with one analogy.
- **Calculate** the small example by hand, step by step.
- **Check units**: s, veh/h, veh/s, LPU, LPU/s, %.

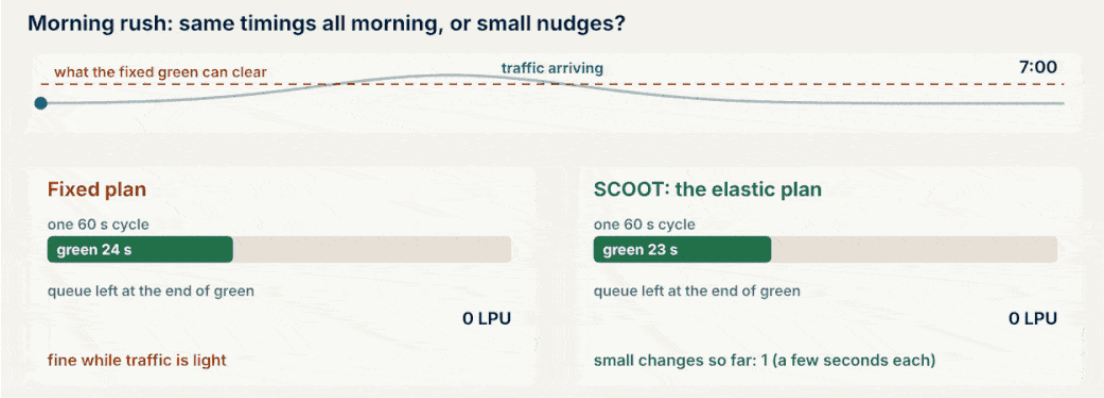
Group	F1	F2	F3	F4	F5	F6	F7	F8
1 Small nudges	●					●	●	●
2 Detectors & LPUs		●	●					
3 Platoons spread		●	●				●	
4 Queue & DoS	●		●	●	●	●		
5 Validation		●		●		●		
6 SCOOT vs others	●				●			●

● = *revise this building block first*

TRY THE NUMBERS Stuck? Re-read the building block on the chip, then your concept section in these slides.

Group 1 Small nudges

Why does SCOOT change the lights only a few seconds at a time?



WATCH Morning rush: a fixed plan vs SCOOT's small, frequent changes.

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. What are the **split**, the **cycle** and the **offset**? Explain each in one line.

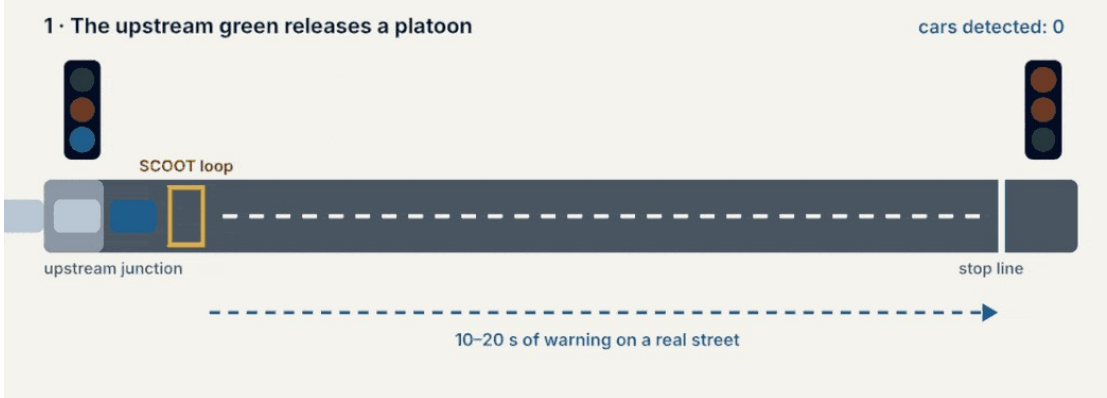
b. Count the timing decisions SCOOT makes at one junction in one hour (60 s cycle, two stages).

c. What could go wrong if SCOOT made one big change instead of many small ones?

Fundamentals: F1 · F6 · F7 · F8
Look in: Section 1 · website: What SCOOT is

Group 2 Detectors & LPUs

Why is SCOOT's detector far up the road, and why does a bus count more than a car?



WATCH A platoon is counted just after the upstream junction, long before the stop line.

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. Where is the SCOOT loop, and what does SCOOT gain from putting it there?

b. What is an LPU? Write the rule used in this lab.

c. Work out the LPUs for a car (4.5 m) and a bus (12 m) at 13.5 m/s over a 2 m loop.

Fundamentals: F2 · F3
Look in: Section 2 · website: Detectors and LPUs

Group 3 Platoons spread

A platoon passes the detector in 10 s. Why does it reach the stop line spread over about 25 s?



WATCH Fast drivers pull ahead, slow drivers fall behind on the way to the stop line.

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. What is a **cyclic flow profile**?

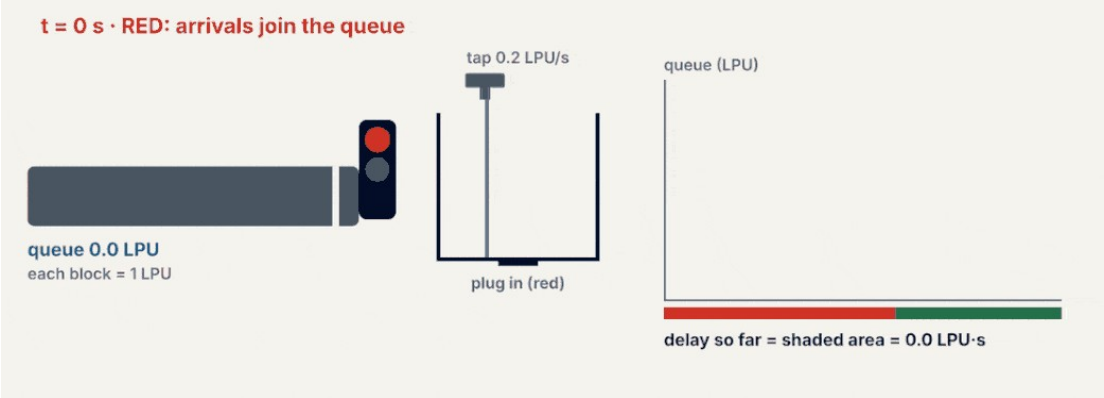
b. Why do the first cars arrive after 0.8 × cruise time, not the full cruise time?

c. What does **F = 0.263** mean? What happens to F on a longer road?

Fundamentals: F2 · F3 · F7
Look in: Section 3 · website: Profiles and dispersion

Group 4 Queue & DoS

SCOOT never sees the queue. How does it know whether a green is long enough?



WATCH The bathtub: red fills the queue, green drains it.

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. Explain the queue model with a bathtub: what are the tap, the plug and the plughole?

b. Arrivals 0.2 LPU/s, red 35 s, green 25 s, SATO 0.6 LPU/s: how big is the queue at the end of red, and when does it clear?

c. Work out the degree of saturation (DoS). Why aim for about 90%, not 100%?

Fundamentals: F1 · F3 · F4 · F5 · F6
Look in: Section 4 · website: The queue model

Group 5 Validation

The model says the queue clears in 32 s; the engineer's stopwatch says 36 s. Who is right, and what do you change?



WATCH A queue drains across the stop line at the saturation occupancy (SATO).

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. What is **SATO**, and why can it differ from street to street?

b. Queue 19.5 LPU (10 cars), start lag 3 s, observed clear time 36 s: what should SATO be?

c. Name two other settings an engineer checks when validating a link.

Fundamentals: F2 · F4 · F6
Look in: Section 5 · website: Validation

Group 6 SCOOT vs others

Is SCOOT always the best way to run traffic lights?



WATCH One junction, the same arrivals, three ways to run the lights (Lab F3).

REPORT BACK one-sentence answer · one key number · explain the animation

INVESTIGATE

a. How do fixed time, actuated and SCOOT signals each decide the green?

b. Compare the mean delay of the three controllers in Lab F3. By how much does SCOOT win?

c. When might a fixed plan or simple actuation be good enough?

Fundamentals: F1 · F5 · F8
Look in: Section 6 · website: Live junction

1 What SCOOT is: the elastic plan

IN PLAIN WORDS

Think of a thermostat that nudges the heating a little every few minutes, rather than a timer that switches between 'winter' and 'summer' settings. Each nudge is small, so one wrong guess cannot do much harm, and the many nudges add up to follow the weather closely.

- Split, Cycle, Offset: three timing knobs
- Small, frequent changes: the elastic plan
- Central computer, a model of every link
- Three speeds: stage change, cycle, 5 minutes

~192

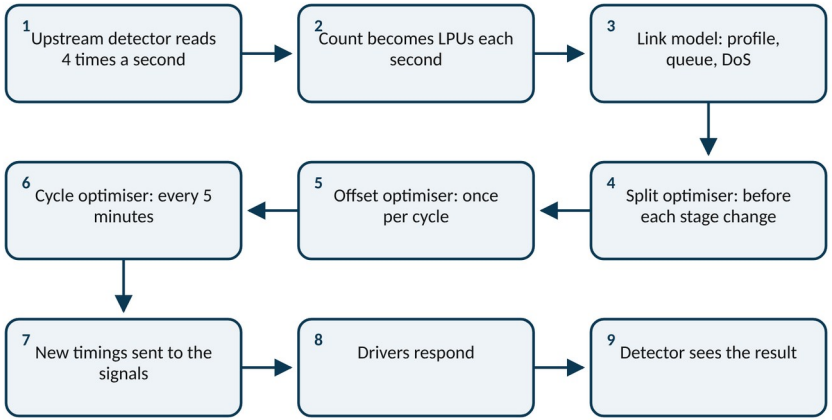
small timing decisions per hour at one junction

Worked example: How many decisions in one hour?

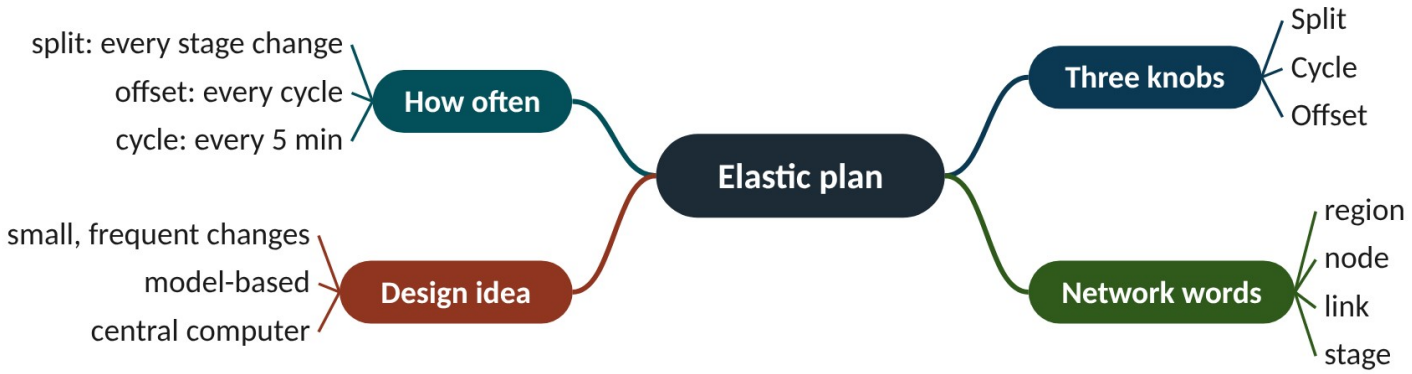
One junction, two stages, a 60 s cycle, in a region whose cycle is reviewed every 5 minutes.

1. Cycles per hour = 3600 ÷ 60 = **60**.
2. Split decisions: one before each stage change, two changes per cycle, so 60 × 2 = **120**.
3. Offset decisions: one per cycle, so **60**.
4. Cycle decisions: one per 5 minutes, so 60 ÷ 5 = **12**.

About 192 small decisions an hour at one junction. That is why each one can be small: there are plenty more to come.



Remember it: Elastic plan



PICTURE THIS · STOP 1: TOWN GATE

At the town gate a giant rubber band is stretched around the traffic light. Three dials on the pole are labelled S, C and O. A tiny engineer twangs the band every few seconds; it stretches, never snaps.

TRAP

SCOOT does not pick a new plan from a library of plans (that is plan selection, an older idea). It edits the plan it is already running, a few seconds at a time.

2 Detectors and link profile units

IN PLAIN WORDS

Imagine a doorman who charges every person one coin to enter, plus two coins for every second they block the door. A quick walker pays a little over one coin; someone pushing a wide trolley pays three. The bill measures how much of the doorway each person used, not just how many came in.

- Loops at the upstream end of each link
- Gives 10-20 seconds of warning
- LPU = vehicles + occupancy
- Long or slow vehicles count more

1.96 vs 3.07

LPUs for a car and a bus at 13.5 m/s

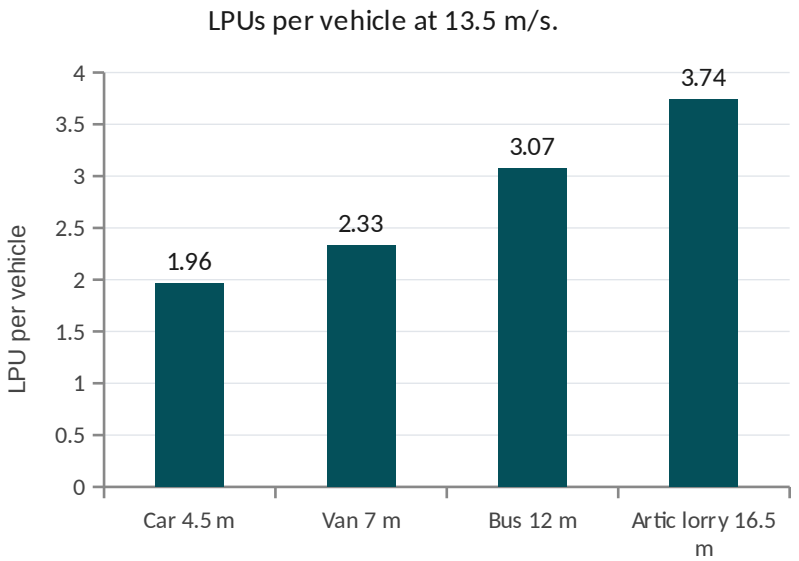
Worked example: A car and a bus at 13.5 m/s over a 2 m loop

occupancy time = (vehicle length + loop length) ÷ speed
LPU = 1 × vehicles + 2 × seconds occupied

Car length 4.5 m. Bus length 12 m. Speed 13.5 m/s (about 49 km/h).

1. Car occupancy: (4.5 + 2) ÷ 13.5 = 6.5 ÷ 13.5 = **0.48 s**.
2. Car LPU: 1 + 2 × 0.48 = **1.96 LPU**.
3. Bus occupancy: (12 + 2) ÷ 13.5 = 14 ÷ 13.5 = **1.04 s**.
4. Bus LPU: 1 + 2 × 1.04 = **3.07 LPU**.
5. Ratio: 3.07 ÷ 1.96 = **1.57**.

The bus counts as about one and a half cars, without the loop ever 'knowing' it was a bus. The SUMO validation lab measured 1.95 LPU per vehicle on average, very close to our car.



Remember it: Detectors & LPUs



PICTURE THIS · STOP 2: TOLL DOORMAT

Just past the gate, a doormat across the whole road is a cash register. Every vehicle pays ONE gold coin to roll on, then TWO coins for every second it sits there. A long red bus groans, paying three coins and dropping one on its foot.

TRAP

Do not think LPUs are 'vehicles'. A queue of 10 cars is about 20 LPUs in this lab. Always check whether a number is in vehicles or LPUs before you compare it.

3 Profiles and platoon dispersion

IN PLAIN WORDS

Picture a school class leaving the gate together and walking to the bus stop. The fast walkers pull ahead, the slow ones fall behind. At the bus stop the class arrives over a longer time than it took to leave, and the busiest moment is less busy.

- Cyclic flow profile: flow by second of the cycle
- Shifted by about 0.8 × cruise time
- Spread by Robertson's model
- Same total, flatter and later peak

F = 0.263

smoothing factor on a 10-second link

Worked example: A 10-second platoon on a 10-second link

$$q_{stop}(t + \beta T) = F \cdot q_{det}(t) + (1 - F) \cdot q_{stop}(t + \beta T - 1)$$
$$F = 1 \div (1 + \alpha\, \beta\, T), \quad \alpha = 0.35,\ \beta = 0.8$$

A platoon crosses the detector at 1 LPU per second for 10 seconds. Cruise time T = 10 s.

1. Lag βT = 0.8 × 10 = **8 s**: the first arrivals appear 8 s after the first detection.

2. F = 1 ÷ (1 + 0.35 × 0.8 × 10) = 1 ÷ 3.8 = **0.263**.

3. Second 1 at the stop line: 0.263 × 1 + 0.737 × 0 = **0.263**.

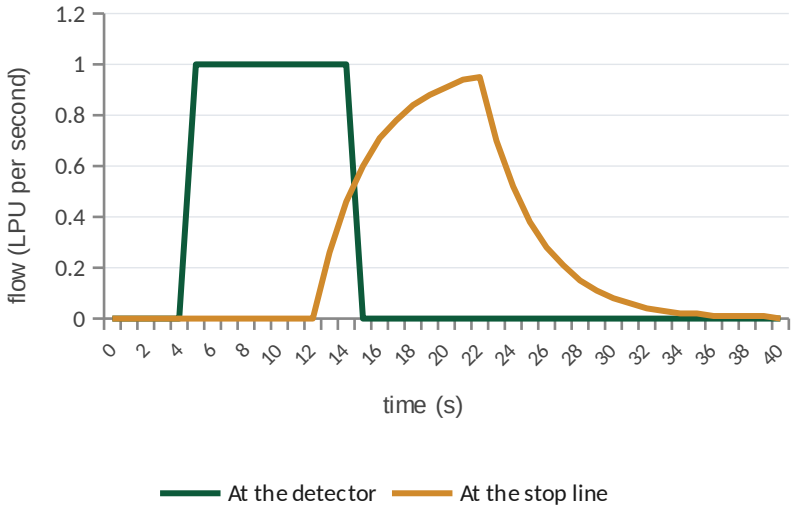
4. Second 2: 0.263 × 1 + 0.737 × 0.263 = **0.457**.

5. Second 3: 0.263 × 1 + 0.737 × 0.457 = **0.600**.

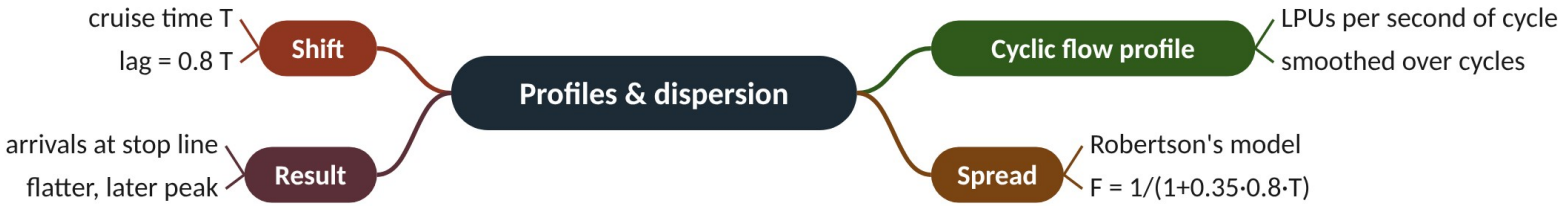
6. Keep going: the arrivals reach a peak of 0.95 after 10 s and then fade away over roughly 15 more seconds.

The same 10 LPUs arrive, but spread over about 25 s with a lower peak (0.95 instead of 1.0). Nothing is lost; it is only rearranged in time.

The same platoon at the detector and at the stop line.



Remember it: Profiles & dispersion



PICTURE THIS · STOP 3: MARCHING-BAND BRIDGE

On a bridge, a marching band of 35 players leaves in a tight block. A sign says 'walk at 80% speed'. By the far end of the bridge the trombones are way ahead and the drummers trail far behind, still playing.

TRAP

The lag is not the full cruise time. In Robertson's model the platoon's front arrives at $\beta T = 0.8\ T$, because the fastest drivers get there early. The rest arrive later in the tail.

4 The queue model and degree of saturation

IN PLAIN WORDS

A queue is a bathtub. Arrivals are the tap, always running. During red the plug is in, so the water rises. During green the plug is out and the water drains at a fixed speed (the saturation occupancy). If the tap is slower than the drain, the tub empties during green.

- Red fills the queue, green drains it at SATO
- Delay is the area under the queue
- DoS = demand ÷ what green can serve
- Aim for 90% or less; PI = delay + K·stops

90%

SCOOT's usual degree-of-saturation target

Worked example: One 60 s cycle on one link

DoS = (LPU arriving per cycle + residual queue) ÷ (effective green × SATO)
PI = delay + K × stops

Arrivals 0.2 LPU/s, steady. Effective red 35 s, effective green 25 s. SATO 0.6 LPU/s. No queue at the start.

1. Queue at the end of red: 0.2 × 35 = **7 LPU**.

2. In green the queue shrinks at 0.6 − 0.2 = 0.4 LPU/s, so it clears after 7 ÷ 0.4 = **17.5 s**.

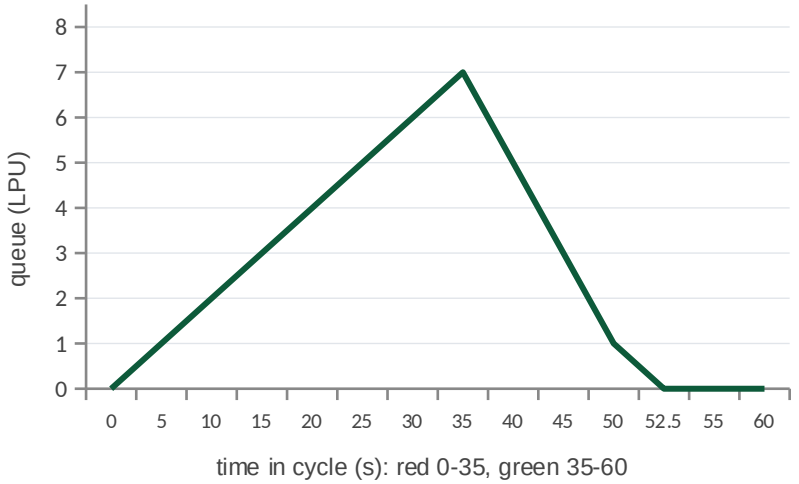
3. Delay = triangle areas: ½ × 35 × 7 + ½ × 17.5 × 7 = 122.5 + 61.25 = **183.75 LPU·s** per cycle.

4. Stops = everyone who arrived before the queue cleared: 0.2 × (35 + 17.5) = **10.5 LPU**.

5. Demand per cycle = 0.2 × 60 = 12 LPU. Capacity = 25 × 0.6 = 15 LPU. DoS = 12 ÷ 15 = **0.80**.

DoS is 80%, under the 90% target, so this link has some spare green to give away.

The modelled queue over one cycle.



Remember it: Queue & DoS



PICTURE THIS · STOP 4: BATHTUB AT THE STOP LINE

At the stop line stands a huge bathtub with a tap that never stops. When the light is red, a rubber plug shaped like a red light is in. On green the plug pops and water swirls out. A gauge on the side is painted red above the 90% mark.

TRAP

A DoS of 100% does not mean the road is jammed forever; it means the green is exactly used up. Above 100% a residual queue carries into the next cycle and grows. Also: DoS uses effective green, not the displayed green.

5 Validation: making the model match the street

IN PLAIN WORDS

Validation is like tuning a guitar by ear. The model is the guitar; the street is the tuning fork. You pluck, listen, and turn the peg until they agree. A perfectly built guitar still plays wrong notes if nobody tunes it.

- The model is only as good as its parameters
- Engineer compares model with the street
- Queue clears early in model → SATO too high
- Lab: SATO 0.67 / 0.89, lags 3 s / 4 s

4 s

clear-time error from a 12% SATO error

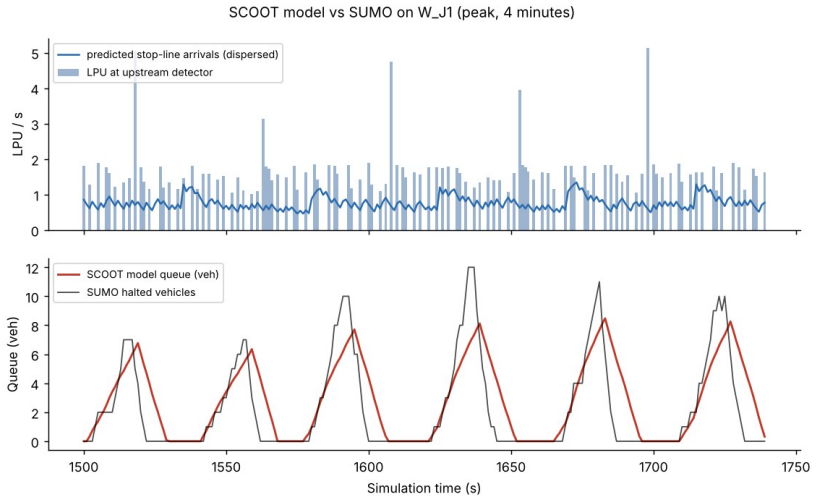
Worked example: Fixing SATO from a stopwatch

queue clear time = start lag + queue ÷ SATO
SATO = queue ÷ (observed clear time − start lag)

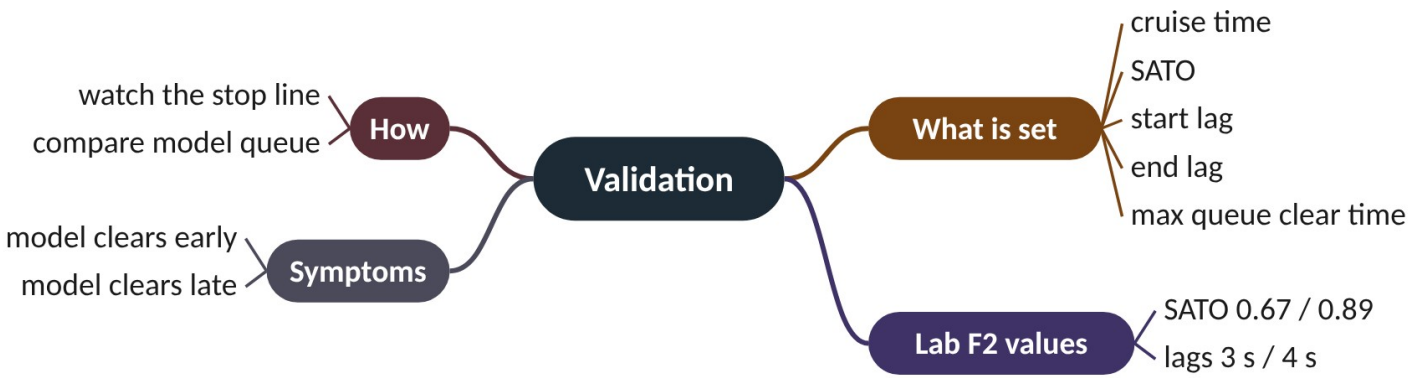
A queue of 10 cars (about 19.5 LPU) is waiting. Model SATO 0.67 LPU/s, start lag 3 s. The engineer sees the queue clear 36 s after green starts.

1. Model clear time: 3 + 19.5 ÷ 0.67 = 3 + 29.1 = **32.1 s**.
2. Observed: **36 s**. The model clears about 4 s early, so its SATO is too high.
3. Corrected SATO = 19.5 ÷ (36 − 3) = 19.5 ÷ 33 = **0.59 LPU/s**.
4. Check: 3 + 19.5 ÷ 0.59 = 3 + 33.0 = **36.0 s**. It now matches.

A 12% change in SATO moved the model's clear time by 4 s. Small parameter errors give big errors in delay and DoS.



Remember it: Validation



PICTURE THIS · STOP 5: GUITAR SHOP

Beside the stop line a guitar shop has a giant guitar strung across the road. An engineer with a stopwatch in her teeth twists a tuning peg marked SATO. The peg has two smaller pegs labelled 3 and 4.

TRAP

Do not 'fix' a model by changing the target DoS or the optimiser. If the queue picture is wrong, every decision built on it is wrong. Validate the model first.

6 SCOOT against fixed time and actuation

IN PLAIN WORDS

Fixed time is a printed bus timetable. Actuation is a driver who leaves as soon as nobody else is running for the bus. SCOOT is a dispatcher with a radio who hears that a crowd is coming in 15 seconds and plans for it.

- Fixed time: a stored plan (Webster, TRANSYT)
- Actuated: reacts to gaps at the stop line
- SCOOT: predicts from upstream, adapts gently
- Lab F3: 19.0 vs 14.5 vs 12.8 s delay

12.8 s

SCOOT mean delay in Lab F3 (fixed 19.0 s)

Worked example: Webster for a two-stage junction

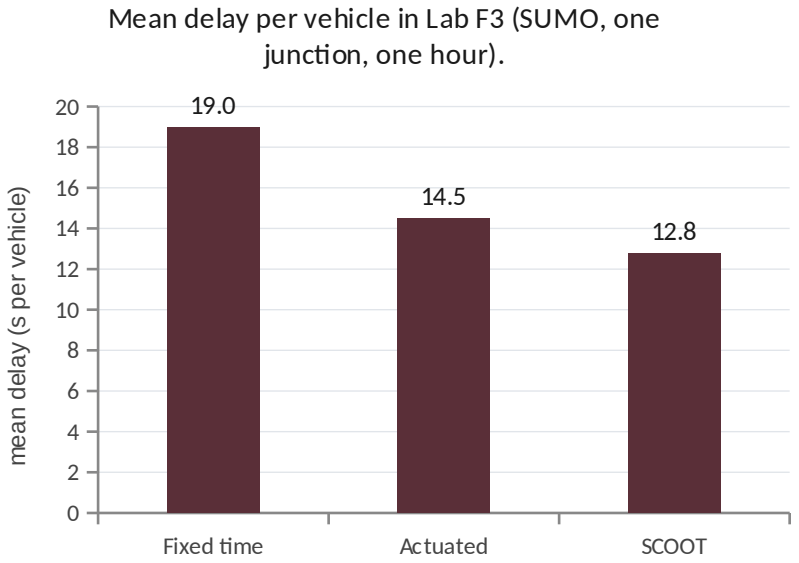
$$C_0 = (1.5\,L + 5) \div (1 - Y)$$

$$g_i = (C_0 - L) \times y_i \div Y$$

Lost time L = 10 s. Main road flow ratio y_A = 0.35, side road y_B = 0.25.

1. Y = 0.35 + 0.25 = **0.60**.
2. C_0 = (1.5 × 10 + 5) ÷ (1 − 0.60) = 20 ÷ 0.40 = **50 s**.
3. Effective green to share = 50 − 10 = **40 s**.
4. Main road: 40 × 0.35 ÷ 0.60 = **23.3 s**. Side road: 40 × 0.25 ÷ 0.60 = **16.7 s**.

A good fixed plan for these flows: 50 s cycle, 23.3 s and 16.7 s of effective green. It stays good only while the flows stay the same, which is why SCOOT adapts.



Remember it: SCOOT vs others



PICTURE THIS · STOP 6: FINISH LINE AT THE CONTROL CENTRE

Outside the control centre three runners finish a race. A robot with a printed timetable on its chest crosses at 19, a jumpy rabbit at 14 and a half, and a calm dispatcher wearing headphones at just under 13. On the podium sits a set of scales labelled '1.5 L + 5'.

TRAP

Webster's formula blows up as Y approaches 1: at Y = 0.9 and L = 10, C0 = 200 s. Real controllers cap the cycle, and when Y ≥ 1 the junction is simply over capacity: no cycle length can fix it.

Report back and discuss

Each group presents its answer in three minutes. Then we compare with the answer slide and the class asks one question.

Group 1 answer Small nudges, many times an hour

Why does SCOOT change the lights only a few seconds at a time?

› **Split** = how the green is shared between stages. **Cycle** = time to run all stages once. **Offset** = timing between neighbouring junctions.

› SCOOT tries a small change before every stage change (split), once per cycle (offset) and every 5 minutes (cycle).

› Decisions per hour: 120 split + 60 offset + 12 cycle = **about 192**.

› Each nudge is a few seconds, so one wrong guess cannot do much harm; together they follow the traffic, like a thermostat.



≈ 192

timing decisions per hour at one junction

DISCUSS

Ask: why not just pick the best plan once each morning? (Traffic changes minute by minute; a stored plan is out of date by the time the peak arrives.)

Group 2 answer Early warning, and a fair measure of traffic

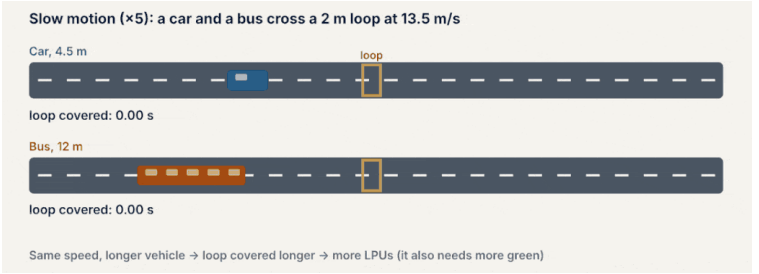
Why is SCOOT's detector far up the road, and why does a bus count more than a car?

› The loop sits just after the **upstream** junction, so SCOOT learns about cars **10–20 s** before they reach the stop line and can get the green ready.

› An **LPU** (link profile unit) mixes a count with how long the loop is covered: LPU = 1 × vehicle + 2 × seconds covered.

› Car: (4.5 + 2) ÷ 13.5 = 0.48 s → **1.96 LPU**. Bus: (12 + 2) ÷ 13.5 = 1.04 s → **3.07 LPU**.

› Long or slow vehicles cover the loop longer and need more green, so they score more.



1.96 vs 3.07

LPUs for a car and a bus

DISCUSS

Ask: what happens to a car's LPU if it crawls over the loop in a jam? (Covered longer, more LPUs: an early sign of congestion.)

Group 3 answer Shift, then smear

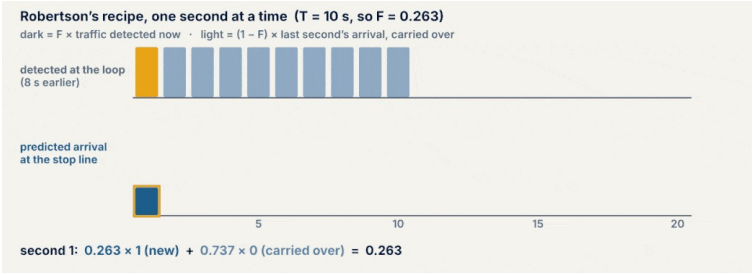
A platoon passes the detector in 10 s. Why does it reach the stop line spread over about 25 s?

› A cyclic flow profile is the LPUs seen in each second of the cycle, averaged over several cycles.

› Shift: the fastest drivers lead, so arrivals start at 0.8 × T = **8 s** (T = 10 s cruise time).

› Smear: each second, **26%** of the new traffic arrives and **74%** of last second's arrivals carry over (F = 1 ÷ (1 + 0.35 × 0.8 × 10) = 0.263).

› Same 10 LPU, lower peak (0.95), about 25 s long. Longer road → smaller F (T = 20 s gives 0.15) → more spread.



F = 0.263

share of each second's traffic that arrives first

DISCUSS

Ask: on a long road, does coordinating the two signals still help much? (Less: the platoon has smeared into steady flow.)

Group 4 answer A model queue, drained by the green

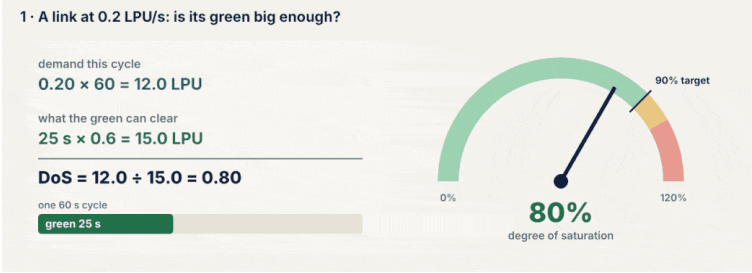
SCOOT never sees the queue. How does it know whether a green is long enough?

› SCOOT **models** the queue from the upstream counts: arrivals are the tap, red is the plug, and SATO is the size of the plughole.

› End of red: 0.2 × 35 = **7 LPU**. In green it shrinks at 0.6 − 0.2 = 0.4 LPU/s, so it clears after 7 ÷ 0.4 = **17.5 s**.

› DoS = demand ÷ what the green can clear = (0.2 × 60) ÷ (25 × 0.6) = 12 ÷ 15 = **0.80**.

› Aiming for about 90% leaves a margin for random surges. Above 100% the queue grows every cycle.



DoS 0.80

under the 90% target: spare green

DISCUSS

Ask: if this link has spare green, where should SCOOT give it? (To the busiest stage at the junction, a few seconds at a time.)

Group 5 answer The street is right: tune the model

The model says the queue clears in 32 s; the engineer's stopwatch says 36 s. Who is right, and what do you change?

› **SATO** is how fast a standing queue drains in green, in LPU per second. It depends on lane width, gradient, turning traffic and parking.

› The street is always right. SATO = queue ÷ (clear time − start lag) = 19.5 ÷ (36 − 3) = **0.59 LPU/s** (the model had 0.67).

› Other settings: journey (cruise) time, start and end lags, and the maximum queue clear time.

› Like tuning a guitar: a well-built model still plays wrong notes until it is tuned to the street.



0.59 LPU/s

SATO after tuning (was 0.67)

DISCUSS

Ask: what goes wrong if SATO is set too high? (The model thinks queues clear sooner, so SCOOT gives too little green and real queues grow.)

Group 6 answer Usually better, not always

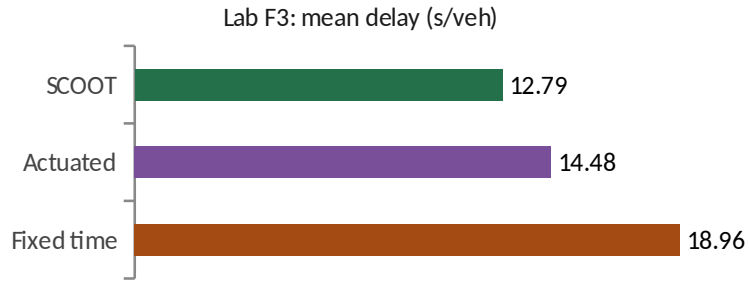
Is SCOOT always the best way to run traffic lights?

› **Fixed time** follows a stored plan. **Actuated** extends the green while cars keep arriving at the stop line. **SCOOT** predicts from upstream and coordinates junctions.

› Lab F3, one junction: 18.96 / 14.48 / **12.79 s** mean delay, so SCOOT has 33% less delay than fixed time and 12% less than actuation.

› Not always: in the four-junction corridor run, actuation had slightly less delay (29.1 vs 30.4 s), while SCOOT had fewer stops and much less bus delay.

› A simple controller can be enough at isolated, lightly used junctions. SCOOT needs working detectors and a validated model.



12.79 s

SCOOT mean delay in Lab F3 (fixed 18.96)

DISCUSS

Ask: which number matters most to you as an engineer: delay, stops, or bus delay? Why?

Three ways to run a traffic light

	Fixed time	Actuated	SCOOT
Detectors	none	at the stop line	upstream, 10-20 s before the stop line
Decides with	a stored plan	gaps between vehicles	a model of every link
Coordinates junctions?	yes, if designed so	no	yes, through offsets and a shared cycle
Reacts to change	no	every second	small steps: stage change, cycle, 5 min
Lab F3 delay	18.96 s	14.48 s	12.79 s

Things beginners mix up

This…	…is not this	The fact that tells them apart
Vehicles	**LPUs**	LPUs add occupancy, so a car is about 2 LPU and a bus about 3.
Cruise time T	**Lag βT**	The platoon's front arrives at 0.8 T, not T.
Displayed green	**Effective green**	Effective green starts a start lag late and runs an end lag into amber.
DoS 100%	**Gridlock**	100% means the green is just used up; queues grow only above it.
Elastic plan	**Plan selection**	SCOOT edits the running plan; plan selection swaps whole stored plans.

Check yourself (1 of 2)

A 7 m van crosses a 2 m loop at 13.5 m/s. Using this lab's LPU rule, how many LPUs?

A. 1.00

B. 1.67

C. 2.33

D. 3.07

Arrivals 0.25 LPU/s, effective green 30 s, SATO 0.5 LPU/s, cycle 60 s, no residual queue. DoS?

A. 0.50

B. 0.83

C. 1.00

D. 1.20

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

In Lab F3 which controller had the lowest mean delay?

A. Fixed time

B. Actuated

C. SCOOT

D. All the same

Webster with Y = 0.95 and L = 10 gives C_0 = 400 s. What is the right conclusion?

A. Use a 400 s cycle

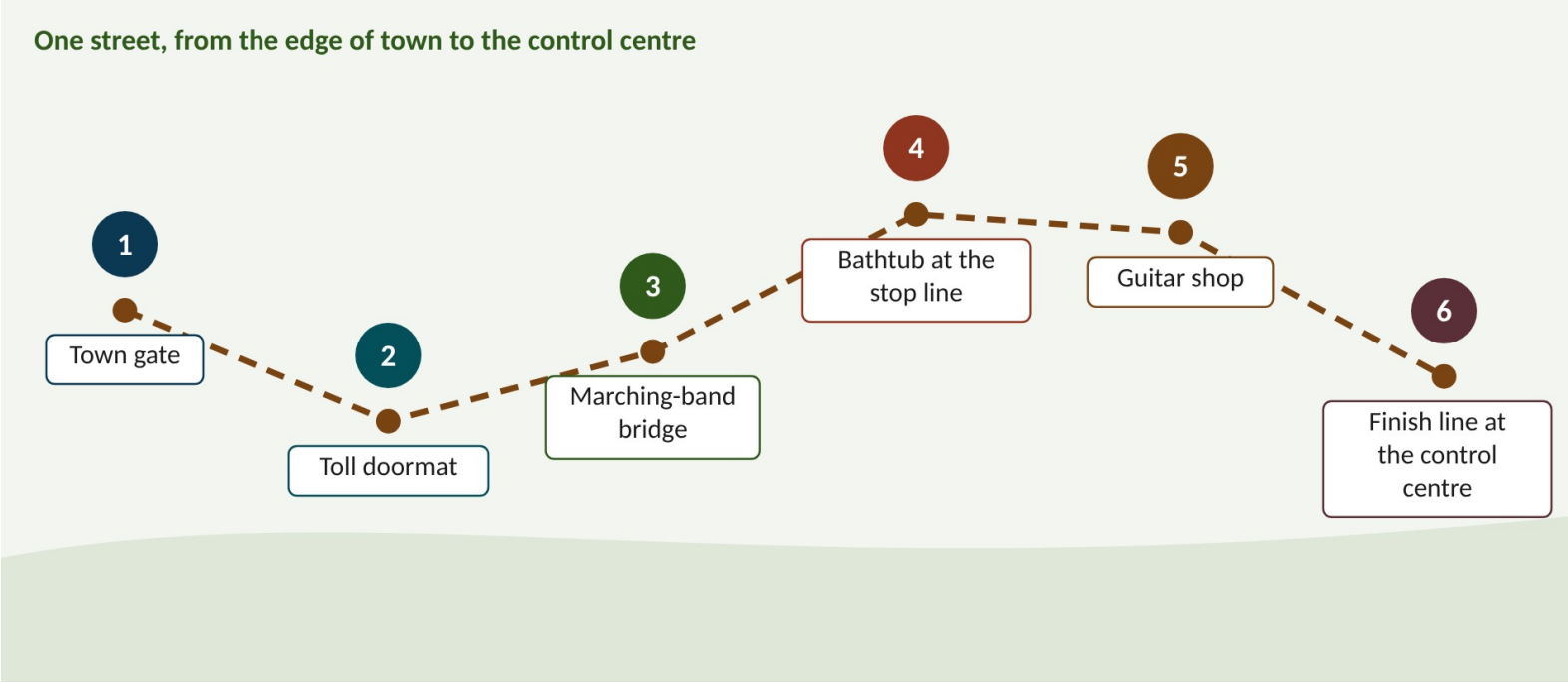
B. The junction is near capacity; cap the cycle and look at capacity, not timing

C. Webster is wrong for all junctions

D. Set Y to 0.6

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Town gate = Elastic plan · 2 Toll doormat = Detectors & LPUs · 3 Marching-band bridge = Profiles & dispersion · 4 Bathtub at the stop line = Queue & DoS · 5 Guitar shop = Validation · 6 Finish line at the control centre = SCOOT vs others

Numbers to remember

- Loop read 4 times a second; 10-20 s warning
- Car ≈ 1.96 LPU, bus ≈ 3.07 LPU (13.5 m/s, 2 m loop)
- $\alpha = 0.35$, $\beta = 0.8$; T = 10 → F = 0.263
- DoS target ≈ 90%
- Lab F2: SATO 0.67 / 0.89 LPU/s per lane, lags 3 / 4 s
- Lab F3 delay: 18.96 / 14.48 / 12.79 s
- Webster: L 10, Y 0.6 → C_0 = 50 s

Keep going

Study guide: SCOOT_Foundation_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scoot-learning-lab